

Physics of Crystalline Materials
MSAE 4200
Worksheet: Group Theory

Names and Roles of team members:

Goal: The goals of this spreadsheet are:

- To develop the basic tools for understanding the important fluctuation-dissipation theorem
- To understand the fluctuation-dissipation theorem

Start this worksheet in class in your groups and discuss it. There will also be some class discussion to get things off to a good start. This worksheet will be finished in groups for homework.

OK, here we go:

Usual preamble:

1. Figure out your group
2. Figure out who is team leader
3. Introduce yourselves to each other
4. Figure out who is scribe for this worksheet
5. Discuss and write down the team goal
6. We will work on this worksheet in groups in class but you will finish up by yourself for homework.

Review of where we are

- We learned that if we have linear problems there are some rather general results that are true, in particular, linear problem + causality mean that we can derive a definite relationship between the real and imaginary parts of the susceptibility.
- This means that if we can measure either the real-part, or the imaginary part, over a sufficiently wide range of frequencies, we can determine the one we don't know from the other using the Kramers-Kronig relations.
- The imaginary part gives the dissipation in the system

(a)